MA3632 Lecture 1 – Data Acquisition and the Data Mining Pipeline

This module is concerned with extracting knowledge and structure from large, heterogeneous datasets using statistical and computational methods. Before any modelling can begin, however, one must address a more basic question: where does the data come from, and in what form does it arrive? This first lecture sets out the framework for the whole module by introducing the *data mining pipeline* and examining its opening stage, *data acquisition*, in some detail. We also establish the vocabulary of data types and data quality that will recur throughout the course.

One point is worth making at the outset. Data preparation – acquisition, cleaning, and feature engineering – often accounts for a substantial proportion of the effort in any real data analysis project. Understanding why this is requires a precise account of what can go wrong at each stage, and that is where we begin.

1 The Data Mining Pipeline

Data mining is not a single operation but a sequence of interdependent stages. We adopt the following decomposition, which we will work through over the course of the module.

Definition 1.1. The **data mining pipeline** is the ordered sequence of transformations that converts raw data into usable knowledge. It consists of the following stages:

(i) **Data acquisition.** Identifying, accessing, and assembling the raw data.

(ii) **Data preparation.** Cleaning, integrating, and transforming the data into a form suitable for analysis. This includes handling missing values, detecting outliers, encoding categorical variables, and constructing new features.

(iii) **Modelling.** Applying statistical or machine learning algorithms to learn patterns, relationships, or structure from the prepared data.

(iv) **Evaluation.** Assessing the quality and generalisability of the learned model using held-out data or cross-validation.

(v) **Deployment and monitoring.** Integrating the model into an operational setting and tracking its behaviour over time.

Remark 1.2. The pipeline is not strictly linear in practice. Findings at the modelling or evaluation stage frequently require returning to earlier steps: a poorly performing model may expose a flaw in the feature construction, which in turn may reveal a problem in the original data acquisition. In practice one should think of the pipeline as a feedback loop.

Remark 1.3. Several industry frameworks formalise this pipeline. The most widely cited is **CRISP-DM** (Cross-Industry Standard Process for Data Mining; Chapman et al., 2000), which identifies six phases: business understanding, data understanding, data preparation, modelling, evaluation, and deployment. The decomposition above focuses on the technical stages of the process; stages (i) and (ii) correspond roughly to CRISP-DM's data understanding and data preparation phases, while CRISP-DM additionally opens with a business understanding phase concerned with project objectives and success criteria rather than the data itself.

2 Data Types and Structures

Before reasoning carefully about data one needs a precise classification of the kinds of values variables can take. This classification determines which operations are mathematically valid and which algorithms are applicable.

Definition 2.1. A **variable** (or *feature*) is a measurable property of an observation. Variables are classified by their *measurement scale*:

(i) **Nominal.** Categories with no intrinsic ordering. Arithmetic on nominal variables is not meaningful. *Examples*: country of origin, blood type, file extension.

(ii) **Ordinal.** Categories with a natural ordering, but with no well-defined notion of distance between categories. *Examples*: survey responses (strongly disagree, ..., strongly agree), academic grades.

(iii) **Interval.** Numerical values for which differences are meaningful but ratios are not, because the zero point is arbitrary. *Examples*: temperature in degrees Celsius, calendar year.

(iv) **Ratio.** Numerical values for which both differences and ratios are meaningful, because zero represents a true absence of the quantity. *Examples*: income, height, elapsed time, file size in bytes.

Remark 2.2. In practice the interval/ratio distinction matters less often than the distinction between *categorical* (nominal or ordinal) and *quantitative* (interval or ratio) variables, since most algorithms require one or the other and do not handle both natively. Encoding strategies – one-hot encoding for nominal variables, ordinal encoding for ordered categories – will be covered in the workshops.

Definition 2.3. A **dataset** is a collection of observations. Once any categorical variables have been suitably encoded, it is typically represented as a matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, where n is the number of observations (rows) and p is the number of features (columns). In supervised learning the dataset is augmented by a *target vector* $\mathbf{y} \in \mathbb{R}^n$ (regression) or $\mathbf{y} \in \{1, \ldots, K\}^n$ (K-class classification).

In modern applications data frequently arrives in forms that do not fit neatly into a matrix.

Definition 2.4. We distinguish the following structural categories:

(i) **Structured data.** Data organised according to a well-defined schema with fixed types per column. Relational databases are the canonical example and are immediately representable as $\mathbf{X}$.

(ii) **Semi-structured data.** Data with a partial or self-describing schema that does not fit a rigid tabular form. *Examples*: JSON documents, XML files, log records.

(iii) **Unstructured data.** Data with no predefined schema. *Examples*: free text, images, audio, video. Feature extraction or representation learning is required before standard algorithms can be applied.

(iv) **Graph-structured data.** Data whose natural representation is a set of entities (nodes) and relationships (edges). *Examples*: social networks, knowledge graphs, molecular structures. Graph-analytic methods are introduced later in the module.

3 Data Acquisition

Data acquisition is the process of identifying, locating, and ingesting the raw data that will feed the rest of the pipeline.

Definition 3.1. A **data source** is any system, process, or repository from which observations can be extracted. A **data acquisition strategy** specifies:

(i) the source or sources from which data will be drawn;

(ii) the mechanism of extraction (database query, API call, sensor read, etc.);

(iii) the format in which data will be received and stored;

(iv) any legal, ethical, or licensing constraints governing its use.

Remark 3.2. Point (iv) is frequently underestimated. Data protection legislation (such as the UK GDPR) imposes strict requirements on the collection, storage, and processing of personal data. A project that ignores these constraints may produce technically correct results that cannot lawfully be used.

We now survey the principal source types encountered in practice.

Databases and data warehouses. Relational databases store data in normalised tables and expose it via SQL. A *data warehouse* is a database optimised for analytical queries rather than transactional updates, typically integrating data from multiple operational systems. Extraction is generally reliable and well-structured, but may require careful handling of joins across large tables.

Flat files. CSV, TSV, JSON, and XML files are ubiquitous interchange formats. They are portable and human-readable, but carry no schema enforcement: two CSV files purporting to contain the same data may use different delimiters, character encodings, or conventions for representing missing values.

Application Programming Interfaces (APIs). Many data providers expose their data through HTTP-based APIs, typically returning JSON or XML. Data acquired this way is subject to rate limits, authentication requirements, versioning changes, and potential service discontinuation, all of which must be planned for in a production pipeline.

Web scraping. When no API is available, data may be extracted directly from HTML pages. This is technically feasible but legally complex (terms of service must be consulted), and the resulting pipeline is fragile: any change to the page layout may silently corrupt the extraction.

Sensors and streaming sources. IoT devices, financial trading systems, and social media platforms generate continuous streams of data. Acquiring such data requires infrastructure for stream processing (e.g. Apache Kafka) and raises questions about buffering, event ordering, and late-arriving records that do not arise in batch settings.

Synthetic data. When real data is unavailable – due to privacy constraints, say, or because one wishes to test a pipeline under controlled conditions – data may be generated algorithmically to mimic the statistical properties of a target distribution.

Definition 3.3. Let P be a probability distribution over $\mathbb{R}^p$. The simplest construction of a **synthetic dataset** of size n drawn from P takes $\mathbf{x}_1, \ldots, \mathbf{x}_n$ to be independent random vectors with $\mathbf{x}_i \sim P$; more general constructions introduce dependence between observations, as required for synthetic time series, spatial, or graph-structured data. When P is not specified analytically but is estimated from a real dataset, the resulting synthetic data is said to be *distribution-matched*.

Remark 3.4. Synthetic data is widely used in testing machine learning pipelines, in privacy-preserving data sharing, and in studying algorithmic behaviour when the ground truth is known. Its limitation is that a model trained on synthetic data may fail to capture properties of the real distribution that were not built into the generative model – a form of model misspecification.

4 Data Quality

Acquired data is rarely fit for immediate analysis. The following dimensions provide a working vocabulary for characterising the ways in which raw data may be deficient.

Definition 4.1. The **quality** of a dataset is assessed along the following dimensions:

(i) **Completeness.** The proportion of values that are present (i.e. not missing). A dataset is complete if every entry of $\mathbf{X}$ is observed.

(ii) **Consistency.** The degree to which the data obeys known logical or domain constraints. *Example*: a patient record with age -5 is inconsistent.

(iii) **Accuracy.** The degree to which recorded values correspond to the true values of the quantities being measured.

(iv) **Timeliness.** Whether the data was collected at the appropriate time and remains current. Stale data may have been accurate when recorded but misleading when used for current predictions.

(v) **Uniqueness.** The absence of duplicate records. Duplicate rows can distort distributional estimates and inflate apparent sample sizes.

The most pervasive quality problem in practice is missing data. We set out a formal taxonomy, due to Rubin (1976), that classifies the *mechanism* by which values come to be missing. This distinction has direct implications for how missing data should be handled, as we shall see in Lecture 2.

Definition 4.2. Let $\mathbf{X}$ be the complete data matrix and let $\mathbf{M}$ be the *missingness indicator matrix*, where $M_{ij} = 1$ if X_{ij} is missing and $M_{ij} = 0$ otherwise. We say that data are:

(i) **Missing Completely at Random (MCAR)** if the probability of missingness does not depend on either the observed or the missing values:

$$\mathbb{P}(\mathbf{M} \mid \mathbf{X}) = \mathbb{P}(\mathbf{M}).$$

(ii) **Missing at Random (MAR)** if the probability of missingness depends only on the observed values, not on the missing values themselves:

$$\mathbb{P}(\mathbf{M} \mid \mathbf{X}) = \mathbb{P}(\mathbf{M} \mid \mathbf{X}_{\text{obs}}).$$

(iii) **Missing Not at Random (MNAR)** if neither condition above holds. Informally, the probability of missingness may depend on the missing values themselves, or on other unobserved quantities, even after conditioning on the observed data.

Example 4.3. Consider a medical survey recording patients' income and health status.

If patients are randomly assigned to skip the income question due to a survey software bug, the missingness is MCAR.

If lower-educated patients are more likely to leave the income question blank, but education is itself recorded in the dataset, the missingness is MAR: it depends on an observed variable but not on income itself.

If high earners systematically decline to report their income, the missingness is MNAR: the probability of the value being missing depends on the value itself.

Remark 4.4. The missing data mechanism matters for the validity of the analysis. Under MCAR, complete-case analysis – discarding all rows with any missing entry – yields unbiased estimates for many standard estimands, though at some loss of efficiency; the exact guarantee depends on the estimand and model being used. Under MAR, more careful methods for handling missing data are generally required, but consistent estimation is still possible under appropriate assumptions. Under MNAR, no imputation strategy can fully correct for the bias without additional assumptions about the missingness mechanism itself. These methods are treated in Lecture 2.

5 Data Provenance

A dataset is not merely a matrix of numbers; it carries a history of decisions about what was measured, how, and by whom. This history – the *provenance* of the data – is essential for reproducibility and for an honest assessment of a model's scope and limitations.

Definition 5.1. The **provenance** of a dataset is the record of its origin, transformations, and chain of custody. A minimal provenance record should specify:

(i) the source (organisation, system, or publication) from which the data was obtained;

(ii) the date and method of acquisition;

(iii) any licence or terms of use governing the data;

(iv) a description of any preprocessing applied before the data enters the pipeline;

(v) known limitations, biases, or caveats associated with the data.

Remark 5.2. The importance of data provenance has been formalised in the *datasheets for datasets* framework (Gebru et al., 2021), which proposes a standardised documentation template analogous to the technical datasheets accompanying electronic components. A well-maintained datasheet allows a future user – or auditor – to assess whether the data is appropriate for a given purpose and to identify potential sources of bias arising from how it was collected.

6 Exercises

Exercise 6.1. A company acquires a dataset of customer transactions by merging three sources: a SQL database of online orders, a CSV export from a legacy point-of-sale system, and a JSON feed from a third-party payment processor.

(i) Identify one potential consistency problem that might arise when merging these three sources.

(ii) The merged dataset has 12% of values missing in the `postcode` field. Give one plausible scenario under which this missingness would be MCAR, one under which it would be MAR, and one under which it would be MNAR.

(iii) List three items that should appear in the provenance record for the merged dataset.

Solution. (i) Acceptable answers include: inconsistent representations of the same customer across systems (different `customer_id` formats); conflicting timestamps due to different time zones; duplicate transactions recorded in both the SQL database and the payment processor feed for the same purchase.

(ii) **MCAR**: The legacy POS system randomly failed to record the postcode due to a software bug affecting a random subset of transactions. **MAR**: Customers who paid by cash (with payment method recorded in the data) were not required to provide a postcode, so missingness depends on the observed variable *payment method*. **MNAR**: Customers in rural areas with non-standard postcodes were more likely to leave the field blank, so the probability of missingness depends on the postcode value itself.

(iii) Any three of: the name and version of each source system; the date of extraction; the licence or data-sharing agreement for each source; a description of the merge key and any deduplication steps applied; known limitations of the legacy POS export (missing fields, date range covered, etc.).

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Exercise 6.2. Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ be a data matrix in which each entry is independently missing with probability $\pi \in (0, 1)$, regardless of its value or the values of any other entries.

(i) Under Rubin's taxonomy (Definition 4.2), what is the missingness mechanism? Justify your answer.

(ii) Let C denote the number of complete rows (rows with no missing entries). Show that $\mathbb{E}(C) = n(1 - \pi)^p$.

(iii) For $n = 1000$, $p = 20$, and $\pi = 0.05$, compute $\mathbb{E}(C)$ to two decimal places, and comment on the practical implication for complete-case analysis.

Solution. (i) The mechanism is MCAR. By assumption, $\mathbb{P}(M_{ij} = 1) = \pi$ independently of X_{ij} and of all other entries, which satisfies $\mathbb{P}(\mathbf{M} \mid \mathbf{X}) = \mathbb{P}(\mathbf{M})$ as required by Definition 4.2(i).

(ii) Row i is complete if and only if none of its p entries is missing. Since entries are independent, $\mathbb{P}(\text{row } i \text{ complete}) = (1 - \pi)^p$. Writing $C = \sum_{i=1}^{n} \mathbf{1}[\text{row } i \text{ complete}]$ and applying linearity of expectation gives $\mathbb{E}(C) = n(1 - \pi)^p$.

(iii) $\mathbb{E}(C) = 1000 \times (0.95)^{20} \approx 358.49$. Even with only 5% missingness per feature, complete-case analysis retains fewer than 36% of rows. This illustrates why naive deletion is costly in moderately high-dimensional settings and motivates the imputation methods studied in Lecture 2.

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Further reading

The material in this lecture is foundational and does not follow any single source closely. The following are recommended for supplementary reading.

Lindholm et al. (2022), Chapter 1. A concise introduction to the supervised learning setup and the role of data in the modelling process; accessible as a first reading before the supervised learning block begins.

James et al. (2023), Section 2.1. Covers the statistical learning framework and the input/output structure that underpins the rest of that text; a useful complement to the matrix formulation in Definition 2.3 above.

Rubin (1976). The foundational paper introducing the formal framework for missing-data mechanisms, including MCAR and MAR, used in Definition 4.2.

Gebru et al. (2021). The datasheets for datasets paper; recommended reading in full for anyone working with real data in a professional or research context.

References

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